CHM111: General Chemistry I: Exam 6: ζ
Dr. Deskins

$\ln\left(\frac{P_2}{P_1}\right) = \frac{k}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$ $\Pi = iMRT$ $n\lambda = 2d\cos\theta$ $\Delta H_{bp} = i \cdot m \cdot K_{bp}$

$\frac{15}{20} \Rightarrow$ 75 %

0 3 5 8 10 13 15 18 20 23 25 28 30 33 35 38 40 43 45 48 50 53 55 57 60 63 65 68 70 73 75 78 80 83 85 88 90 93 95 98 100
0 0.5 1 1.5 2 2.5 3 3.5 4 4.5 5 5.5 6 6.5 7 7.5 8 8.5 9 9.5 10 10.5 11 11.5 12 12.5 13 13.5 14 14.5 15 15.5 16 16.5 17 17.5 18 18.5 19 19.5 20

Not 100%, but shows clear reasoning and can benefit other students. See answer key for correct answers and explanation.

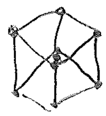
These questions assess your understanding of the material from Chapters 10, 11 of OpenStax Chemistry. Please write your name on the sheets of paper that you use. Carefully read each question. For questions with numerical calculations, you must show each major step necessary to find your final answer. For questions without numerical calculations, you must include clear and correct reasoning to support your answer. Remember to report the units and correct number of significant figures in your answers.

1. A colloid consists of a liquid dispersed in a gas. Name this class of colloid and give an everyday example.

Aerosol, Spray paint

ANSWER: Aerosol, spray paint ✓

2. An ionic compound crystallizes with a simple cubic array of anions and the cation in the cubic (body-center) hole. What is the cation:anion ratio (the formula unit) in the compound? Justify by counting ions per cell.



$8 \cdot \frac{1}{8} = 1$ anion
$1 \cdot 1 = 1$ cation

ANSWER: 1:1 ✓

3. Predict what happens to solubility when a lake is heated by a power plant. Identify whether temperature or pressure is the key factor and explain the observation.

The solubility of gasses decreases, while the solubility for most solids and increases.
Both T and P are factors in gas solubility, but since P is constant, T determines solubility.

ANSWER: Temperature. ✓

4. A solution shows a freezing point depression of 0.512°C when 100. g of water is used as the solvent. How many grams of benzoic acid ($C_7H_6O_2$, $\mathcal{M}$ = 122.12 g/mol) are dissolved? ($K_f(H_2O)$ = 1.86°C kg/mol)

ANSWER: 3.36g ✓
3SF

$\Delta H_f = i \cdot m \cdot k_f$

$m = \frac{\Delta H_f}{i \cdot K_f}$

0.275 mol/L
0.0275 mol/100mL or 100g ≈ 3.36g

5. Classify a hard, high-melting solid of silicon and carbon (carborundum) (SiC) as ionic, metallic, covalent network, or molecular, and predict its melting point (high or low) and electrical conductivity. Justify your answer.

SiC is a covalent network ceramic, meaning strong bonds. This leads to high mp.
Also, no free electrons means SiC is a non-conductor.

ANSWER: Covalent network, high mp, non-Conductive. ✓

6. A 0.500 M solution of NaCl (NaCl) has a density of 1.02 g/mL. Calculate the molality of the solution. ($\mathcal{M}$ = 58.44 g/mol)

ANSWER: 0.500m ✗
(or 0.490m if mol/1kg soln.)

0.500M NaCl = 0.5 moles NaCl/L
molality = moles/1000g solvent ?
molality is not moles/1000g solution, but solvent
(if it was, then molality = 0.490m)

yes
calculate it
(Answer key)

+5

7. A 20.3-g sample of liquid water at 0°C releases 4.92 kJ of heat. Determine the final state and temperature. ($\Delta H_{fus} = 334$ J/g.)

ANSWER: Some ice, some water, 0°C

$4{,}920 J - (20.3 \cdot 334) = -1860.2$

Heat released freezes some of the water, but not all. This means temperature stays at 0°C

8. When sodium hydroxide (NaOH) dissolves in water the solution warms up. A student says this warming alone proves the dissolution is spontaneous. Is that reasoning sound? Explain.

ANSWER: Below

No, dissolution is driven by both enthalpy and entropy changes, and even enthalpy unfavorable rxns can still occur due to ~~entropy~~ (an increase in entropy)

9. How many hydrogen-bond acceptors does one molecule of CH_3OH have?

ANSWER: 2

$H_3C-\ddot{O}-H$ lone pairs/negative regions to accept lone pairs

10. A solution contains a crystal is added and rapid crystallization occurs at a given temperature. Classify the solution as saturated, unsaturated, or supersaturated. You must explain your answer choice for credit.

ANSWER: Supersaturated

If more solute is dissolved than typically possible at a given T, adding a seed crystal causes the excess to precipitate out. This means the soln. is supersaturated

11. Calculate the heat absorbed when 35.4-g of water (H_2O) melts completely at its melting point of 0°C. $\Delta H_{fus} = 6.01$ kJ/mol.

ANSWER: 11.8 kJ

$\frac{35.4 g}{18.02 g/mol} = 1.965 mol H_2O \cdot \frac{6.01 kJ}{mol} = 11.81 kJ$ 3SF

12. On a heating curve for water, the boiling plateau is much longer than the melting plateau. What does this tell you? Justify.

ANSWER: $\Delta H_{evap} > \Delta H_{fus}$

Having a longer plateau means that more (heat) energy is absorbed before the next phase change occurs. This tells us water has a comparatively higher ΔH_{evap} compared to ΔH_{fusion}.

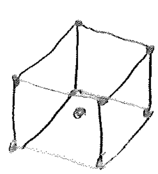
13. For sulfur dioxide (SO_2): normal melting point −75°C, normal boiling point −10°C, triple point (−75°C, 0.017 atm), critical point (158°C, 78 atm). What phase is present at −101°C and 1 atm? Justify your answer.

ANSWER: Solid.

−101°C is well below SO_2 melting point. Given they are at the same T, SO_2 would be solid

14. What is the coordination number of each atom in a body-centered cubic (BCC) structure?

ANSWER: 8



Each atom is touching 8 other atoms.
Coordination # = 8

+8

15. How many grams of H_2O can freeze when 76.1-kJ of heat is released during freezing at 0°C? Assume $\Delta H_{fus} = 334\,J/g$.

ANSWER: 228 g H_2O 3SF ✓

$$\frac{76{,}100\,J}{334\,J/g} = 227.84\,g\ H_2O$$

16. Predict the shape of the meniscus formed by mercury in a copper tube. Justify using cohesive and adhesive forces.

ANSWER: Upwards at edges (concave?) ✗

(But metallic bonding in Hg is far stronger) Mercury would be attracted to metalic bonding in Cu, and the liquid would slightly rise at the edges, forming a concave meniscus

17. Two solutions are prepared: solution A is a 0.200 m solution of NaCl (NaCl) in water, and solution B is a 0.200 m solution of glucose ($C_6H_{12}O_6$) in water. Which solution has the greater boiling point elevation? Calculate both values. (Assume complete dissociation for ionic solutes.) $K = 0.512\ °C/\text{kg·mol}$

ANSWER: Solution A ✓

Soln. A bp ≈ 102.05

Soln B bp ≈ 101.02

The Van't Hoff factor (i) would nearly double the effectiveness of NaCl compared to glucose.

$\Delta H_{evap} = i \cdot m \cdot k_{evap}$

$\Delta H_{evap} = 1 \cdot 0.200 \cdot 0.512 = 0.1024$

18. A beam of light is passed through gelatin solution? Will the Tyndall effect be observed? Classify the mixture as a true solution, colloid, or suspension. Offer brief reasoning for why you chose your answer. Answers without valid explanation earn no credit.

ANSWER: True soln., no Tyndall effect — colloid. ✗

Gelatin influences water's mp and Physical behavior, but still completely dissolves. Very small particles = no Tyndall effect (Gel)atin

19. An unknown metal crystallizes in a FCC structure with unit cell edge length 361.49 pm and density 8.94 g/cm³. Calculate the molar mass of the metal and identify it.

ANSWER: 1570 g/mol, unknown 3SF ✗

atoms in a $4.724 \times 10^{-29}\,cm^3$

each atom occupies $1.181 \times 10^{-24}\,cm^3$

8.467×10^{23} atoms/cm³

$\frac{1}{2} \cdot 6 = 3$ 4 atoms per cell

$\frac{1}{8} \cdot 8 = 1$ 4 atoms in a $(361.5 \times 10^{-10}\,cm)^3$ area

See answer key

20. Calculate the vapor pressure of a solution prepared by dissolving 36.0 g of sucrose ($C_{12}H_{22}O_{11}$) in 100. g of water at 80°C. The vapor pressure of pure water at 80°C is 355.1 mmHg.

ANSWER: > 355.1 mmHg ✗

always lowers P_{vap} since solute inhibits vaporization at surface

$$\frac{4\ \text{atoms}}{1\ \text{cell}} \left| \frac{1\ \text{cell}}{4.724 \times 10^{-29}\,cm^3} \right| \frac{1\,cm^3}{8.94\,g} \left| \frac{1\ \text{mol}}{6.022 \times 10^{23}\ \text{atoms}} \right. = \frac{1\ \text{mol}}{6.358 \times 10^{-4}\,g}$$

or

1573 g/mol

+2

1. Aerosol; e.g., fog. $\because$ a liquid-in-gas dispersion defines this colloid type.
2. Cation:anion $= 1:1$ $\because$ counting the ions per unit cell from a simple cubic array of anions with the cation in the cubic (body-center) hole gives that ratio (the cesium chloride (CsCl) structure).
3. Dissolved o_2 decreases. $\because$ gas solubility decreases with increasing temperature; fish kills can result.
4. $m = \dfrac{\Delta T}{K_f = 1.86} = \dfrac{0.512}{1.86} = 0.275\,\text{mol/kg}$; $n = 0.275 \times 0.100\,\text{kg} = 0.0275\,\text{mol}$; $m_{benz} = 0.0275\,\text{mol} \times 122.12\,\text{g/mol} = 3.36\,\text{g}$
5. Covalent network solid $\because$ its composition implies extreme hardness, a very high melting point, and poor conduction.
6. Per 1 L of solution: mass solution $= 1.02 \times 10^3\,\text{g}$; mass solute $= 0.500\,\text{mol} \times 58.44\,\text{g/mol} = 29.2\,\text{g}$; mass solvent $= 1.02 \times 10^3\,\text{g} - 29.2\,\text{g} = 996\,\text{g}$; $m = \dfrac{0.500\,\text{mol}}{0.996\,\text{kg}} = 0.502\,\text{mol/kg}$
7. Freezing all 20.3 g would release 6.78 kJ, but only 4.92 kJ is removed $\therefore$ partial freeze: an ice–water mixture at 0°C, with 14.7 g frozen.
8. Warming (exothermic ΔH) favors dissolution but does not by itself prove spontaneity; both the enthalpy change and the entropy change matter. $\because$ dissolving also raises entropy here, both factors are favorable and the salt dissolves spontaneously.
9. 2 $\because$ 1 O–H bond (donor) + 2 lone pairs on O (acceptor).
10. Supersaturated $\because$ excess dissolved solute precipitates on the crystal nucleus.
11. $q = n\Delta H_{\text{fus}} = \dfrac{35.4\,\text{g}}{18.015\,\text{g/mol}} \times 6.01\,\text{kJ/mol} = 11.8\,\text{kJ}$
12. ΔH_{vap} is much larger than ΔH_{fus} $\because$ plateau length is proportional to heat absorbed; vaporization separates molecules completely, while melting only loosens the lattice.
13. Solid $\because$ at 1 atm, T is below the normal melting point (−75°C).
14. 8 $\because$ center atom touches all 8 corner atoms.
15. $m = \dfrac{q}{\Delta H} = \dfrac{76.1 \times 10^3\,\text{J}}{334\,\text{J/g}} = 228\,\text{g}$
16. Convex (bulges up in the middle) $\because$ adhesion of mercury to copper is weaker than cohesion within the mercury, so the liquid pulls away from the walls.
17. Solution A: $\Delta T = iKm = 2 \times 0.512 \times 0.200 = 0.205\,°\text{C}$; Solution B: $\Delta T = 1 \times 0.512 \times 0.200 = 0.102\,°\text{C}$. Solution A has the greater boiling point elevation $\because$ NaCl dissociates into 2 particles ($i = 2$) vs. glucose ($i = 1$).
18. Yes, Tyndall effect is observed. Colloid (gel) $\because$ polymer chains 1–1000 nm scatter light.
19. $M = \dfrac{\rho \cdot N_A \cdot a^3}{Z} = \dfrac{8.94 \times 6.022 \times 10^{23} \times (361.49 \times 10^{-10})^3}{4} = 63.5\,\text{g/mol}$ $\therefore$ the metal is copper (Cu).
20. $\chi_{H_2O} = \dfrac{5.55}{5.55 + 0.105} = 0.981$; $P = 0.981 \times 355.1\,\text{mmHg} = 348\,\text{mmHg}$; $\Delta P = 6.60\,\text{mmHg}$ lowering.